MSG No. 128.410 P_I4/mnc [Type IV, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `4/mm'm'`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,6,9,10,11,14,20,21,23,24,28,29,31,32]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[4,5,7,8,12,13,15,16,17,18,19,22,25,26,27,30]`

Table 2: Wyckoff site: `2b`, site symmetry: `4/mm'm'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,3,6,9,10,11,14,20,21,23,24,28,29,31,32]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[4,5,7,8,12,13,15,16,17,18,19,22,25,26,27,30]`

Table 3: Wyckoff site: `4c`, site symmetry: `mm'm'.`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,6,9,14,20,21,28,29]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[2,3,10,11,23,24,31,32]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[4,5,12,13,17,22,25,30]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[7,8,15,16,18,19,26,27]`

Table 4: Wyckoff site: `4d`, site symmetry: `-4'm'2`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,6,7,8,26,27,28,29]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[2,3,4,5,25,30,31,32]`
3	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,14,15,16,18,19,20,21]`
4	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[10,11,12,13,17,22,23,24]`

Table 5: Wyckoff site: `4e`, site symmetry: `4m'm'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,6,28,29,31,32]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[4,5,7,8,25,26,27,30]`
3	$[0,\ 0,\ -z]$	`[9,10,11,14,20,21,23,24]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[12,13,15,16,17,18,19,22]`

Table 6: Wyckoff site: `8f`, site symmetry: `..2/m'`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,8,25,32]`
2	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[2,4,26,28]`
3	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[3,5,27,29]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[6,7,30,31]
5	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[9,16,17,24]
6	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[10,12,18,20]
7	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[11,13,19,21]
8	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[14,15,22,23]

Table 7: Wyckoff site: 8g, site symmetry: 2m'm'.

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	[1,6,28,29]
2	$\left[\frac{1}{2}, 0, z\right]$	[2,3,31,32]
3	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	[4,5,25,30]
4	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	[7,8,26,27]
5	$\left[0, \frac{1}{2}, -z\right]$	[9,14,20,21]
6	$\left[\frac{1}{2}, 0, -z\right]$	[10,11,23,24]
7	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	[12,13,17,22]
8	$\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$	[15,16,18,19]

Table 8: Wyckoff site: 8h, site symmetry: m.2'm'

No.	position	mapping
1	$[x, x, 0]$	[1,14,23,32]
2	$[-x, x, 0]$	[2,11,21,28]
3	$[x, -x, 0]$	[3,10,20,29]
4	$\left[x + \frac{1}{2}, \frac{1}{2} - x, \frac{1}{2}\right]$	[4,13,19,26]
5	$\left[\frac{1}{2} - x, x + \frac{1}{2}, \frac{1}{2}\right]$	[5,12,18,27]
6	$[-x, -x, 0]$	[6,9,24,31]
7	$\left[x + \frac{1}{2}, x + \frac{1}{2}, \frac{1}{2}\right]$	[7,16,17,30]
8	$\left[\frac{1}{2} - x, \frac{1}{2} - x, \frac{1}{2}\right]$	[8,15,22,25]

Table 9: Wyckoff site: 8i, site symmetry: m2'm'.

No.	position	mapping
1	$[x, 0, 0]$	[1,14,20,29]
2	$[0, x, 0]$	[2,11,23,32]
3	$[0, -x, 0]$	[3,10,24,31]
4	$\left[x + \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[4,13,17,30]
5	$\left[\frac{1}{2} - x, \frac{1}{2}, \frac{1}{2}\right]$	[5,12,22,25]
6	$[-x, 0, 0]$	[6,9,21,28]
7	$\left[\frac{1}{2}, x + \frac{1}{2}, \frac{1}{2}\right]$	[7,16,18,27]
8	$\left[\frac{1}{2}, \frac{1}{2} - x, \frac{1}{2}\right]$	[8,15,19,26]

Table 10: Wyckoff site: `8j`, site symmetry: `m2'm'`.

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,14,20,29]`
2	$\left[\frac{1}{2},\ x,\ 0\right]$	`[2,11,23,32]`
3	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[3,10,24,31]`
4	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[4,13,17,30]`
5	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	`[5,12,22,25]`
6	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[6,9,21,28]`
7	$\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[7,16,18,27]`
8	$\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[8,15,19,26]`

Table 11: Wyckoff site: `16k`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[1,7]`
2	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{4}\right]$	`[2,5]`
3	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	`[3,4]`
4	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[6,8]`
5	$\left[-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[9,15]`
6	$\left[x+\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	`[10,13]`
7	$\left[\frac{1}{2}-x,\ x,\ \frac{3}{4}\right]$	`[11,12]`
8	$\left[x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[14,16]`
9	$\left[x+\frac{1}{2},\ x,\ \frac{3}{4}\right]$	`[17,23]`
10	$\left[-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[18,21]`
11	$\left[x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[19,20]`
12	$\left[\frac{1}{2}-x,\ -x,\ \frac{3}{4}\right]$	`[22,24]`
13	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{4}\right]$	`[25,31]`
14	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[26,29]`
15	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[27,28]`
16	$\left[x+\frac{1}{2},\ x,\ \frac{1}{4}\right]$	`[30,32]`

Table 12: Wyckoff site: `16l`, site symmetry: `m..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,14]`
2	$[-y,\ x,\ 0]$	`[2,11]`
3	$[y,\ -x,\ 0]$	`[3,10]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[4,13]`
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[5,12]`
6	$[-x,\ -y,\ 0]$	`[6,9]`
7	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[7,16]`
8	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[8,15]`
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[17,30]`
10	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[18,27]`
11	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[19,26]`
12	$[x,\ -y,\ 0]$	`[20,29]`

continued ...

Table 12

No.	position	mapping
13	$[-x,\ y,\ 0]$	[21,28]
14	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[22,25]
15	$[y,\ x,\ 0]$	[23,32]
16	$[-y,\ -x,\ 0]$	[24,31]

Table 13: Wyckoff site: 16m, site symmetry: ..m'

No.	position	mapping
1	$[x,\ x,\ z]$	[1,32]
2	$[-x,\ x,\ z]$	[2,28]
3	$[x,\ -x,\ z]$	[3,29]
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[4,26]
5	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[5,27]
6	$[-x,\ -x,\ z]$	[6,31]
7	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[7,30]
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[8,25]
9	$[-x,\ -x,\ -z]$	[9,24]
10	$[x,\ -x,\ -z]$	[10,20]
11	$[-x,\ x,\ -z]$	[11,21]
12	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[12,18]
13	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[13,19]
14	$[x,\ x,\ -z]$	[14,23]
15	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[15,22]
16	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[16,17]

Table 14: Wyckoff site: 16n, site symmetry: .m'.

No.	position	mapping
1	$[0,\ y,\ z]$	[1,28]
2	$[-y,\ 0,\ z]$	[2,31]
3	$[y,\ 0,\ z]$	[3,32]
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[4,25]
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[5,30]
6	$[0,\ -y,\ z]$	[6,29]
7	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[7,26]
8	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[8,27]
9	$[0,\ -y,\ -z]$	[9,20]
10	$[y,\ 0,\ -z]$	[10,23]
11	$[-y,\ 0,\ -z]$	[11,24]
12	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[12,17]
13	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[13,22]
14	$[0,\ y,\ -z]$	[14,21]
15	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[15,18]
16	$\left[y+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[16,19]

Table 15: Wyckoff site: 32o, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-y,\ x,\ z]$	[2]
3	$[y,\ -x,\ z]$	[3]
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[4]
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[5]
6	$[-x,\ -y,\ z]$	[6]
7	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[7]
8	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[8]
9	$[-x,\ -y,\ -z]$	[9]
10	$[y,\ -x,\ -z]$	[10]
11	$[-y,\ x,\ -z]$	[11]
12	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[12]
13	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[13]
14	$[x,\ y,\ -z]$	[14]
15	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[15]
16	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[16]
17	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[17]
18	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[18]
19	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[19]
20	$[x,\ -y,\ -z]$	[20]
21	$[-x,\ y,\ -z]$	[21]
22	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[22]
23	$[y,\ x,\ -z]$	[23]
24	$[-y,\ -x,\ -z]$	[24]
25	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[25]
26	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[26]
27	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[27]
28	$[-x,\ y,\ z]$	[28]
29	$[x,\ -y,\ z]$	[29]
30	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[30]
31	$[-y,\ -x,\ z]$	[31]
32	$[y,\ x,\ z]$	[32]